

RESEARCH ARTICLE

Real-time PM_{2.5} mapping and anomaly detection from AirBoxes in Taiwan

G. Huang¹ | L.-J. Chen² | W.-H. Hwang³ | S. Tzeng⁴ | H.-C. Huang⁵

¹Institute of Statistics, National Tsing Hua University, Hsinchu, Taiwan

²Institute of Information Science, Academia Sinica, Taipei, Taiwan

³Department of Applied Mathematics, National Chung Hsing University, Taichung, Taiwan

⁴Department of Applied Mathematics, National Sun Yat-sen University, Kaohsiung, Taiwan

⁵Institute of Statistical Science, Academia Sinica, Taipei, Taiwan

Correspondence

H.-C. Huang, Institute of Statistical Science, Academia Sinica, Taipei 11529, Taiwan.
Email: hchuang@stat.sinica.edu.tw

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Abstract

Fine particulate matter (PM_{2.5}) has gained increasing attention due to its adverse health effects to human. In Taiwan, it was conventionally monitored by large environmental monitoring stations of the Environmental Protection Administration. However, only a small number of 77 monitoring stations are currently established. Recently, a project using a large number of small devices, called AirBoxes, was launched in March 2016 to monitor PM_{2.5} concentrations. Although thousands of AirBoxes have been deployed across Taiwan to give a broader coverage, they are mostly located in big cities and their measurements are less accurate. In this paper, we apply a robust kriging method that provides a smoothly varied real-time PM_{2.5} concentration map and its associated standard error map. In addition, we develop a novel spatio-temporal control chart that monitors anomalous measurements by utilizing neighboring AirBox information. Our method automatically adapts to different neighboring structures at different AirBox locations without the need to specify a neighborhood range. The proposed method has abilities to detect potential emission sources, malfunctioned AirBoxes, and AirBoxes that are wrongly put indoors.

KEYWORDS

control chart, fine particulate matter, kriging, multiresolution spline basis functions, robust regression, robust variogram estimation, spatial prediction

1 | INTRODUCTION

Air pollution is a growing concern for the public. The health impact of air pollution exposure has been widely recognized since the 1950s. For example, the London smog in December 1952 is estimated to cause more than 3,000 excess deaths compared with previous years (Bell & Davis, 2001). Since then, adverse health effects of air pollution have been widely investigated all over the world. A recent World Health Organization report estimates that outdoor air pollution was responsible for the premature deaths of three million people under the age of 60 in 2012 (World Health Organization, 2016).

Many air pollutants have been reported to have adverse effects on humans, such as ozone (O₃), carbon monoxide (CO), sulfur dioxide (SO₂), nitrogen dioxide (NO₂), and particulate matter (PM), where PM refers to particles found in the air, including dust, dirt, soot, smoke, and liquid droplets, which is often classified according to their sizes into PM₁₀ (coarse particles with diameters less than or equal to 10 micrometers) and PM_{2.5} (fine particles with diameters less than or equal to 2.5 micrometers). Exposure to PM is known to be associated with respiratory disease (Lee, Ferguson, & Mitchell, 2009) and cardiovascular illness (US Environmental Protection Agency, 2011), and these associations are believed to be causal because PM (particularly PM_{2.5}) particles can be inhaled into the thoracic region of the respiratory tract. A study of the global burden of disease in 1990–2010 ranks ambient PM_{2.5} concentrations ninth out of all health risk factors (Lim et al., 2012). Clearly, PM_{2.5} is coming more and more into focus in environmental monitoring and environmental health.

PM_{2.5} is conventionally monitored by large ground monitoring stations or remote sensing devices. Because levels of PM_{2.5} vary widely in space and time, various methods have been developed to estimate PM_{2.5} levels for locations with no monitoring station nearby. For example, Liu, Sarnat, Kilaru, Jacob, and Koutrakis (2005) and Lee, Liu, Coull, Schwartz, and Koutrakis (2011) incorporated aerosol optical depth data from satellites in a linear regression model and a linear mixed model, respectively, to obtain daily ground-level PM_{2.5} concentrations. A comprehensive review on estimating ground-level PM_{2.5} concentrations based on aerosol optical depth can be found in the work of Chu et al. (2016). Many spatial and spatio-temporal kriging models have also been developed for mapping PM_{2.5}. Calder (2008) introduced a Bayesian dynamic convolution model, which not only utilizes data from PM_{2.5} monitoring stations but also extracts information from data observed in a larger PM₁₀ monitoring network. Paciorek, Yanosky, Puett, Laden, and Suh (2009) considered an additive space–time model to estimate monthly PM₁₀ and PM_{2.5}. Datta, Banerjee, Finley, Hamm, and Schaap (2016) developed a dynamic nearest neighbor Gaussian process model that incorporates numerical outputs from chemistry transport models. Nevertheless, all these methods consider estimating PM_{2.5} on a daily, weekly, or even monthly basis due to that they do not have data at fine spatial and temporal resolutions.

PM_{2.5} monitoring and estimation are also essential to exposure estimation and can be used to study the health impacts of PM_{2.5} by linking personal PM_{2.5} exposures to diseases. For example, Dominici et al. (2016) used PM_{2.5} data in the Air Quality System database and found their associations with cardiovascular and respiratory hospital admissions. Pannullo et al. (2017) used PM_{2.5} data on a 12 km × 12 km regular grid in England and studied the impact of PM_{2.5} on respiratory diseases.

In Taiwan, a network of 66 air quality monitoring stations was established by the Environmental Protection Administration (EPA) in 1994 (Hung et al., 2012). It has been extended to 77 monitoring stations so far. This monitoring network provides valuable information of Taiwan air quality and makes many PM_{2.5}-related research problems possible during the last two decades. For example, Hung et al. (2012) studied the risk of death from breast cancer in Taiwan using PM_{2.5} as a proxy marker. Chen and Yang (2017) investigated the association between PM_{2.5} and hospital admissions for chest pain in Taipei. However, these studies are somewhat hindered by not having good personal PM_{2.5} exposure estimates because the network is too sparse to give a complete coverage of PM_{2.5} at a fine spatial resolution in Taiwan.

Recently, a project that combines Internet of Things (IoT) to PM_{2.5} monitoring was launched in March 2016. The project plans to deploy a large number of small, low-cost devices, called AirBoxes, across Taiwan to provide real-time PM_{2.5} information at high spatial and temporal resolutions. So far, more than two thousand AirBoxes have been deployed. It is designed as a participatory ecosystem, which not only enables people to access real-time data on Google Maps but also provides immediate alerts when PM_{2.5} levels are dangerous. However, measurements from AirBoxes are not as accurate as those from large monitoring stations. In addition, they are not professionally deployed. For example, they may be put indoors or close to small-scale emission sources, such as restaurants or temples. A comprehensive introduction to this AirBox sensor network can be found in Chen et al. (2017).

To ensure data quality for managing a large network sensing system, anomaly detection is crucial. Anomaly detection is an extremely challenging task for IoT systems, especially when the sensing devices are deployed in uncontrolled environments (Zhang, Meratnia, & Havinga, 2010). There have been several studies of anomaly detection in large systems, and they can be classified into three categories based on their fundamental algorithms.

1. *Clustering-based methods*

These methods arrange similar data into clusters. Anomalous instances either lie in sparse and small clusters or are away from normal cluster centroids determined by a threshold. While these methods do not require prior knowledge of the data distribution in the sensor readings and can adapt to new data instances over time (Zhang et al., 2010), it is challenging to determine the threshold of a cluster width that separates anomalies from normal instances. In addition, data obtained from nonhomogeneous environments cannot be accommodated easily (Moshtaghi, Rajasegarar, Leckie, & Karunasekera, 2009). Consequently, this type of methods is generally used as quick filters for detecting outliers (e.g., Haque, Rahman, & Aziz, 2015; Hayes & Capretz, 2015). They are not suitable for anomaly detection for large sensor networks.

2. *Machine-learning methods*

Many machine-learning methods have been applied to detect anomalies in a sensor network. For example, Murphree (2016) applied neural networks, and Ayadi, Zouinkhi, Boussaid, and Abdelkrim (2015) compared three machine-learning algorithms for outlier detection: an active learning method, a density-based method, and feature bagging. In general, this kind of methods requires a significant amount of labeled training data to cover all types of sensor readings in the system. Additionally, their performances rely on choosing good tuning parameters, which in some situations may be difficult or computationally intensive.

3. Statistical methods

Wu et al. (2007) considered a spatial outlier detection method that incorporates neighboring information. However, their method relies on a threshold, which is selected by an ad hoc procedure with little justification. Paschalidis and Chen (2010) developed a statistical framework that considers both spatial and temporal correlations among sensor readings for anomaly detection. However, their framework relies on long periods of anomaly-free observations, which may not be possible for many environmental monitoring applications. In addition, the detection process is relatively long and cannot be implemented in real time. Recently, Chen et al. (2017) developed an anomaly detection framework as an IoT-based real-time PM_{2.5} monitoring system. The framework is based on simple streaming data processing, and it conducts real-time anomaly detection using a set of heuristic rules and threshold values obtained from historical data. However, the performance of the framework is dependent on the selection of neighborhoods, which is subjective and yet to be evaluated.

In this paper, we aim to achieve the following two goals. First, we would like to obtain real-time PM_{2.5} concentration at any location in Taiwan with its estimation error bar. We adopt the kriging approach. To account for complex topographical features in Taiwan and outlying measurements from AirBoxes, we consider a spatial regression model with a class of multiresolution spline basis functions (Tzeng & Huang, 2018) as regressors and apply a robust method for parameter estimation. This provides precise PM_{2.5} estimation at any location in Taiwan even when there are no AirBoxes nearby. Our second goal is to develop a spatio-temporal control chart that can automatically monitor anomalous measurements by utilizing neighboring AirBox information. This allows us to detect potential emission sources, malfunctioned AirBoxes, and AirBoxes that are put indoors within a statistical framework. Note that some people wrongly put their AirBoxes indoors to monitor air quality at their homes, which tend to report much lower concentrations particularly when air purifiers are in operation. Identifying those indoor AirBoxes is essential for data management and can avoid producing misleading results in subsequent data analysis.

The remainder of this paper is organized as follows. In Section 2, we describe the AirBox data set used in this paper. Section 3 introduces our statistical methodologies, including the proposed kriging (i.e., spatial prediction) method to estimate PM_{2.5} and a novel spatio-temporal control chart to detect anomalies. An application of our methods to the AirBox data set is provided in Section 4. Finally, a brief discussion is given in Section 5.

2 | AIRBOX DATA

The study region in this paper is Taiwan, an island located in the Asia-Pacific region, 160 km southeast of Mainland China. Taiwan is a long and narrow island stretching from north to south with about 395 km long and 144 km in width. The majority of Taiwan is covered by mountains. The highest point is Yushan Peak at 3,952 m. Most mountain ranges run from north to south with 268 summits over 3,000 m, which mostly block the transport of air pollution from the west to the east. The population in Taiwan is approximately 23 million with 70 percent concentrated in the five western metropolitan areas (Taipei, Taoyuan, Taichung, Tainan, and Kaohsiung), where plains and basins are located. Figure 1 shows a population density map of Taiwan.

The PM_{2.5} data we analyzed are from 1,283 AirBoxes across Taiwan from January 1, 2017 to February 28, 2017, which can be obtained from the Open Data Portal provided by the LASS community at <https://pm25.lass-net.org>. The sample rate is about 5–6 min, and about 30% of data are missing due to that some AirBoxes were deployed after January 1, 2017, and some encountered occasional disconnection problems to the Internet. Due to some privacy concern, the locations of the AirBoxes are only given in degrees latitude and longitude to the third decimal place. Therefore, the precision is about 111 m in the north–south direction and 102 m in the east–west direction around Taiwan.

We first aggregate the data into hourly data at each data location using the simple average, resulting in 1,416 potential measurements for each location. Figure 2a shows all the AirBox measurements at 8:00 on January 26, 2017. The AirBox locations can be seen to cluster at a few big cities in the west. In general, the PM_{2.5} concentrations vary smoothly in space. Nevertheless, from Figure 2a, we see a few unusual measurements that are either much higher or lower than nearby measurements. For example, there are a few red spots surrounded by yellow and green points in North Taiwan. Those with too large values may be close to some emission sources (e.g., restaurants or factories). On the other hand, those with too small values (green spots surrounded by red points) are likely due to AirBoxes being put indoors.

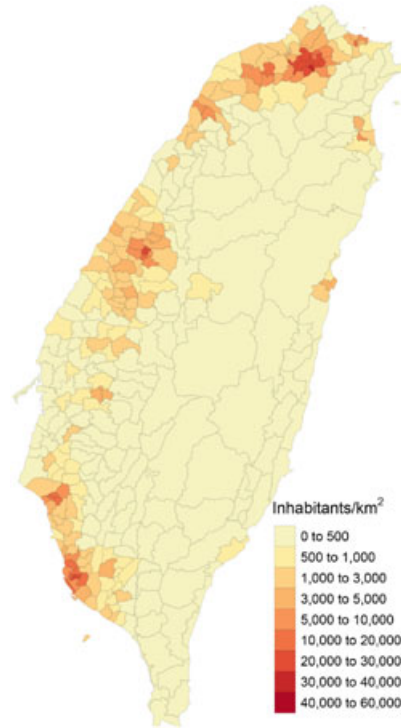


FIGURE 1 A population density map of Taiwan

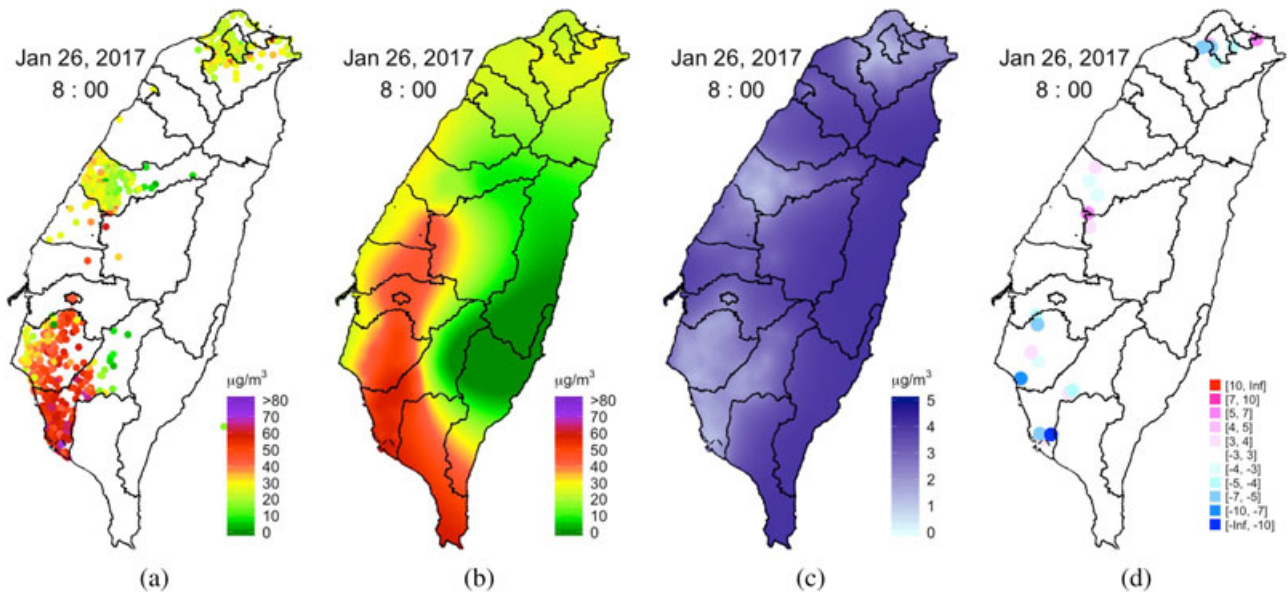


FIGURE 2 An illustration of the proposed kriging method. (a) $PM_{2.5}$ AirBox measurements at 8:00 on January 26, 2017. (b) Estimated $PM_{2.5}$ map by the proposed kriging based on the data in (a). (c) The corresponding estimated standard error map of (b). (d) The anomalies identified by the standardized residuals. $PM_{2.5}$ = fine particulate matter

3 | METHODOLOGIES

3.1 | Mapping $PM_{2.5}$ by kriging with robust model fitting

We shall provide a $PM_{2.5}$ map and its associated standard error map for each time separately by fitting a separate model. We treat the underlying $PM_{2.5}$ levels in space as a spatial process $\{y(s) : s \in D\}$ for each time (i.e., hour), where $D \subset \mathbb{R}^2$

corresponds to a region containing Taiwan, and for notational simplicity, we suppress the dependence of $y(\mathbf{s})$ on time. At each time, the process is assumed to have the following decomposition:

$$y(\mathbf{s}) = \boldsymbol{\beta}' \boldsymbol{\phi}(\mathbf{s}) + \eta(\mathbf{s}) = \sum_{k=1}^K \beta_k \phi_k(\mathbf{s}) + \eta(\mathbf{s}); \quad \mathbf{s} \in D, \quad (1)$$

where $\boldsymbol{\phi}(\mathbf{s}) = (\phi_1(\mathbf{s}), \dots, \phi_K(\mathbf{s}))'$ are the first K multiresolution spline basis functions of Tzeng and Huang (2018), $\phi_1(\mathbf{s}) \equiv 1$, $\boldsymbol{\beta} = (\beta_1, \dots, \beta_K)'$ are regression coefficients, and $\eta(\cdot)$ is a zero-mean spatial process with an exponential covariance function $C(\mathbf{s}, \mathbf{u}) = \text{cov}(y(\mathbf{s}), y(\mathbf{u})) = v_y^2 \exp(-\|\mathbf{s} - \mathbf{u}\|/\lambda)$ to capture the spatial dependencies of $\text{PM}_{2.5}$. Here $v_y^2 = \text{var}(y(\mathbf{s})) = \text{var}(\eta(\mathbf{s}))$ and λ is a spatial dependence parameter. Suppose that we observe data $\mathbf{z} \equiv (z(\mathbf{s}_1), \dots, z(\mathbf{s}_n))'$ at n distinct locations, $\mathbf{s}_1, \dots, \mathbf{s}_n \in D$, with additive white noise according to

$$\mathbf{z} = \mathbf{y} + \boldsymbol{\varepsilon}, \quad (2)$$

where $\mathbf{y} = (y(\mathbf{s}_1), \dots, y(\mathbf{s}_n))'$, $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, v_\varepsilon^2 \mathbf{I})$ is uncorrelated with $\mathbf{y}$, and $\mathbf{I}$ is the identity matrix of order n . We would like to estimate $y(\mathbf{s}_0)$ at any location $\mathbf{s}_0 \in D$ based on $\mathbf{z}$ regardless of whether $\mathbf{s}_0$ is close to some AirBoxes. Note that the first K basis functions of Tzeng and Huang (2018) depend on $\{\mathbf{s}_1, \dots, \mathbf{s}_n\}$, but not on $\mathbf{z}$, which can be used to obtain a rank- K approximation to the thin-plate spline surface (Wood, 2003).

As there may be outlying measurements from AirBoxes, we apply a robust method for model fitting. First, we estimate $\boldsymbol{\beta}$ using Huber's M-estimator (see Huber & Ronchetti, 2009) as follows:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^K} \sum_{i=1}^n \rho\left(\frac{e_i}{\hat{\sigma}}\right),$$

where $e_i = z(\mathbf{s}_i) - \boldsymbol{\beta}' \boldsymbol{\phi}(\mathbf{s}_i)$,

$$\rho(x) = \begin{cases} x^2/2, & \text{if } |x| \leq c, \\ c|x| - c^2/2, & \text{if } |x| > c, \end{cases}$$

and $\hat{\sigma} = \text{median}_i(e_i - \text{median}_j(e_j))/0.6745$. Here, we choose $c = 1.345$, which gives an efficiency of 95% if the regression errors are normally distributed. Next, we estimate the spatial covariance parameters $\boldsymbol{\theta} = (v_y^2, \lambda, v_\varepsilon^2)'$ by fitting the exponential variogram model using weighted least squares (Cressie, 1985) based on variograms at several spatial lags up to 100 km estimated from the robust method of Cressie and Hawkins (1980). Plugging in the resulting estimates $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\theta}} = (\hat{v}_y^2, \hat{\lambda}, \hat{v}_\varepsilon^2)'$ to the simple kriging (i.e., optimal linear prediction) formula of $y(\mathbf{s}_0)$, we obtain

$$\hat{y}(\mathbf{s}_0) = \hat{\boldsymbol{\beta}}' \boldsymbol{\phi}(\mathbf{s}_0) + \mathbf{c}(\mathbf{s}_0; \hat{\boldsymbol{\theta}})' (\boldsymbol{\Sigma}(\hat{\boldsymbol{\theta}}) + \hat{v}_\varepsilon^2 \mathbf{I})^{-1} \left(\mathbf{z} - \left(\hat{\boldsymbol{\beta}}' \boldsymbol{\phi}(\mathbf{s}_1), \dots, \hat{\boldsymbol{\beta}}' \boldsymbol{\phi}(\mathbf{s}_n) \right)' \right), \quad (3)$$

where $\mathbf{c}(\mathbf{s}_0; \boldsymbol{\theta}) = v_y^2 (\exp(-\|\mathbf{s}_1 - \mathbf{s}_0\|/\lambda), \dots, \exp(-\|\mathbf{s}_n - \mathbf{s}_0\|/\lambda))'$, and $\boldsymbol{\Sigma}(\boldsymbol{\theta}) = \text{var}(\mathbf{y})$ is an $n \times n$ matrix with the (i, j) th entry $v_y^2 \exp(-\|\mathbf{s}_i - \mathbf{s}_j\|/\lambda)$. The corresponding standard error $\text{var}(\hat{y}(\mathbf{s}_0) - y(\mathbf{s}_0))^{1/2}$ after plugging in $\hat{\boldsymbol{\theta}}$ is given by

$$\left\{ \hat{v}_y^2 - \mathbf{c}(\mathbf{s}_0; \hat{\boldsymbol{\theta}})' (\boldsymbol{\Sigma}(\hat{\boldsymbol{\theta}}) + \hat{v}_\varepsilon^2 \mathbf{I})^{-1} \mathbf{c}(\mathbf{s}_0; \hat{\boldsymbol{\theta}}) \right\}^{1/2}. \quad (4)$$

3.2 | Anomaly detection

The kriging map obtained in the previous subsection can be used as a baseline to detect anomalies. Let $\{\hat{y}_t(\mathbf{s}) : \mathbf{s} \in D\}$ be the estimated $\text{PM}_{2.5}$ map obtained from (3) at time t based on $\mathbf{z}_t = (z_t(\mathbf{s}_1), \dots, z_t(\mathbf{s}_n))'$ observed at locations $\{\mathbf{s}_1, \dots, \mathbf{s}_n\}$, where $t = 1, \dots, T$. We propose to monitor anomalies using the standardized residuals as follows:

$$r_t(\mathbf{s}_i) = \frac{z_t(\mathbf{s}_i) - \hat{y}_t(\mathbf{s}_i)}{\hat{\sigma}_t(\mathbf{s}_i)}; \quad i = 1, \dots, n, t = 1, \dots, T, \quad (5)$$

where $\hat{\sigma}_t^2(\mathbf{s}_i)$ is an estimate of

$$\sigma_t^2(\mathbf{s}_i) = \text{var}(z_t(\mathbf{s}_i) - \hat{y}_t(\mathbf{s}_i)) = v_\varepsilon^2 + v_y^2 - \mathbf{c}(\mathbf{s}_i; \boldsymbol{\theta})' (\boldsymbol{\Sigma}(\boldsymbol{\theta}) + \sigma_\varepsilon^2 \mathbf{I})^{-1} \mathbf{c}(\mathbf{s}_i; \boldsymbol{\theta})$$

after substituting $\hat{\boldsymbol{\theta}} = (\hat{v}_y^2, \hat{\lambda}, \hat{v}_\varepsilon^2)'$ for $\boldsymbol{\theta} = (v_y^2, \lambda, v_\varepsilon^2)'$ in $\sigma_t^2(\mathbf{s}_i)$.

Note that $r_t(\mathbf{s}_i)$ has a standard normal distribution if the parameters σ_y^2 , λ , and σ_ε^2 are known. A high positive value of $r_t(\mathbf{s}_i)$ indicates that $z_t(\mathbf{s}_i)$ is higher than its neighboring observations. In contrast, a low negative value of $r_t(\mathbf{s}_i)$ suggests that $z_t(\mathbf{s}_i)$ is comparatively lower. Different from the work of Chen et al. (2017), we do not need to specify a specific neighborhood range because it is automatically taken care of by the proposed kriging method.

For each AirBox, we plot its control chart, $r_t(\mathbf{s}_i)$; $i = 1, \dots, T$, and monitor whether it is under control without having excessive large or small values. We follow the common practice by setting plus and minus three standard deviations as the upper and lower control limits for our control charts. In other words, the AirBox at location $\mathbf{s}_i$ and time t is considered to be abnormal if $|r_t(\mathbf{s}_i)| > 3$ because $P(|r_t(\mathbf{s}_i)| > 3) \approx 0.3\%$ if the device is normal.

Other than real-time monitoring, we can also use $\{r_t(\mathbf{s}_i)\}$ to identify emission sources, or unusual AirBoxes, including malfunctioned AirBoxes or those that are put indoors. We rank the potential unusual AirBoxes according to the root mean squared error (RMSE),

$$\text{RMSE}_i = \left\{ \frac{1}{|T_i|} \sum_{t \in T_i} (r_t(\mathbf{s}_i))^2 \right\}^{1/2}, \quad (6)$$

where $T_i = \{t : z_t(\mathbf{s}_i) \text{ is observed}\}$ and $|T_i|$ denotes the number of elements in T_i . The ones with large RMSE_i values indicate that the AirBoxes tend to produce outlying observations. We further decompose RMSE_i in terms of squared bias b_i^2 and variance V_i as follows:

$$\begin{aligned} \text{RMSE}_i &= (b_i^2 + V_i)^{1/2}, \\ b_i &= \frac{1}{|T_i|} \sum_{t \in T_i} r_t(\mathbf{s}_i), \\ V_i &= \frac{1}{|T_i|} \sum_{t \in T_i} \left\{ r_t(\mathbf{s}_i) - \frac{1}{|T_i|} \sum_{t \in T_i} r_t(\mathbf{s}_i) \right\}^2. \end{aligned} \quad (7)$$

We classify the AirBoxes with high RMSEs into three groups separated by the upper and lower control limits according to $b_i \geq 3$, $b_i \in (-3, 3)$ and $b_i \leq -3$. An AirBox with $b_i \geq 3$ implies that its average standardized residual value is above the upper control limit. That is, the AirBox tends to produce significantly higher $\text{PM}_{2.5}$ measurements than its neighboring AirBoxes. Consequently, the AirBox is more likely to locate close to an emission source. An AirBox with $b_i \leq -3$ indicates that its average standardized residual value falls below the lower control limit by having many $\text{PM}_{2.5}$ measurements significantly lower than those from neighboring AirBoxes. This could happen when the AirBox is put indoors. Finally, an AirBox with high RMSE_i but with $b_i \in (-3, 3)$ tends to have a high variation such that its $\text{PM}_{2.5}$ measurements fall above the upper control limit on some occasions, while they fall below the lower control limit on others. This may indicate that the AirBox is malfunctioned or is located in a more complex environment.

4 | AIRBOX DATA ANALYSIS RESULTS

4.1 | Kriging maps of $\text{PM}_{2.5}$

We applied the proposed kriging method developed in Section 3.1 to AirBox data at each hour from January 1, 2017, to February 28, 2017, where we chose $K = 25$ basis functions, which explains about 60% of variation and makes the resulting spatial process $\eta(\cdot)$ in (1) closer to stationary. Precise selection of K is not very important because if the underlying $\text{PM}_{2.5}$ process is not captured by the mean, it can still be captured by the spatially dependent process $\eta(\cdot)$. Figure 3 shows the second to 25th basis functions based on the data locations observed at 8:00 on January 26, 2017, which are ordered in terms of their degrees of smoothness. These functions have a unique property that they allow more structures for locations with more data around and tend to behave like linear surfaces for areas with no data nearby. Therefore, they avoid artifacts that may be produced by other types of basis functions. Figure 4 shows the time series plots for the estimators $\hat{v}_\varepsilon^2$ and $\hat{v}_y^2/\hat{\lambda}$ of v_ε^2 and v_y^2/λ , which as shown by Zhang and Zimmerman (2005), are the two key parameters for spatial prediction using the exponential covariance model with a nugget effect. Both time series show daily periodic patterns with peaks at frequency 1/24 in their periodograms.

Figure 2b shows the estimated $\text{PM}_{2.5}$ map at 8:00 on January 26, 2017, obtained from the proposed kriging method, where Western, particularly Southwestern, Taiwan can be seen to have much higher $\text{PM}_{2.5}$ concentrations than Eastern Taiwan. This is typically the case due to heavier traffic and a few thermal power plants located in Western Taiwan. It is also

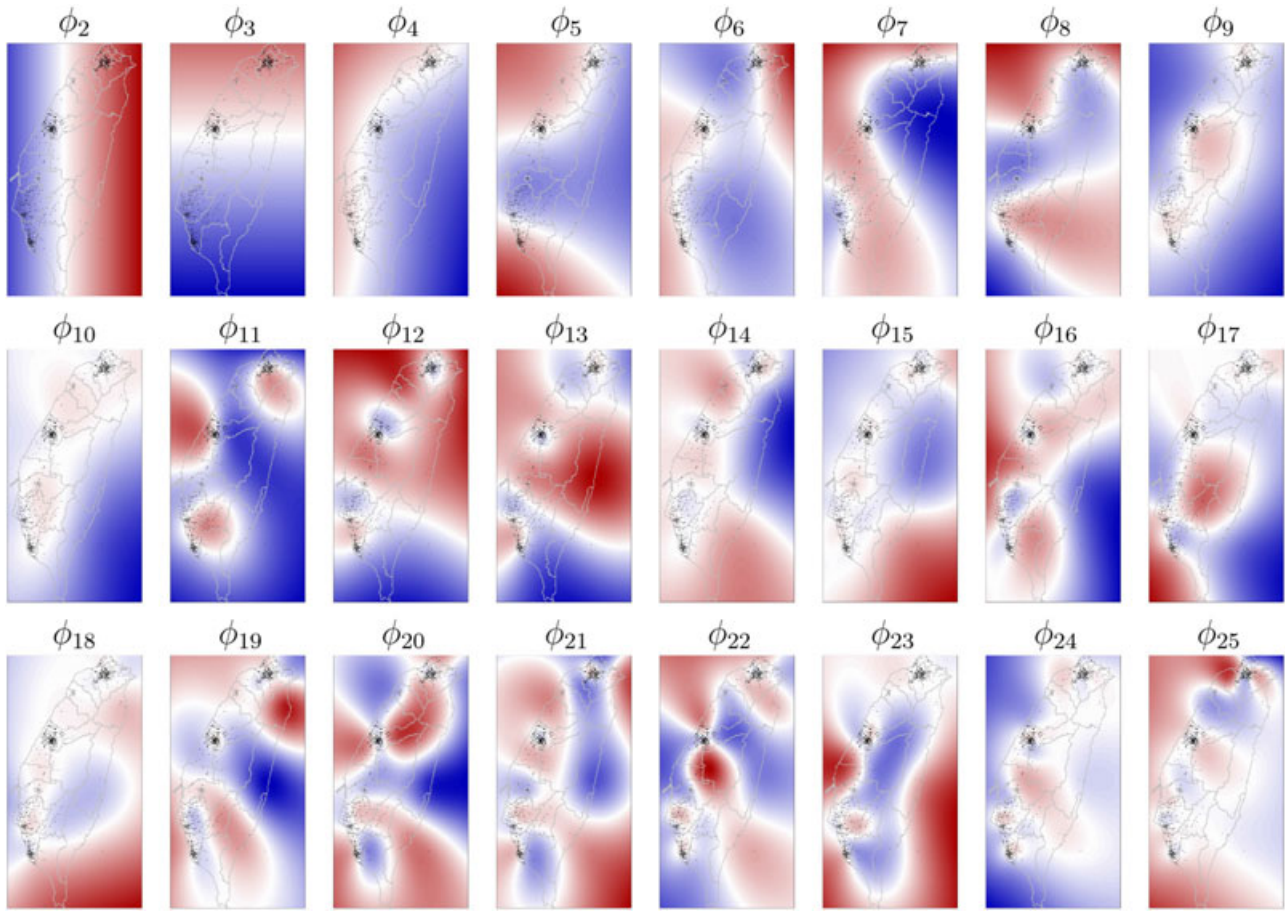


FIGURE 3 Multiresolution spline basis functions: $\phi_2(\cdot), \dots, \phi_{25}(\cdot)$

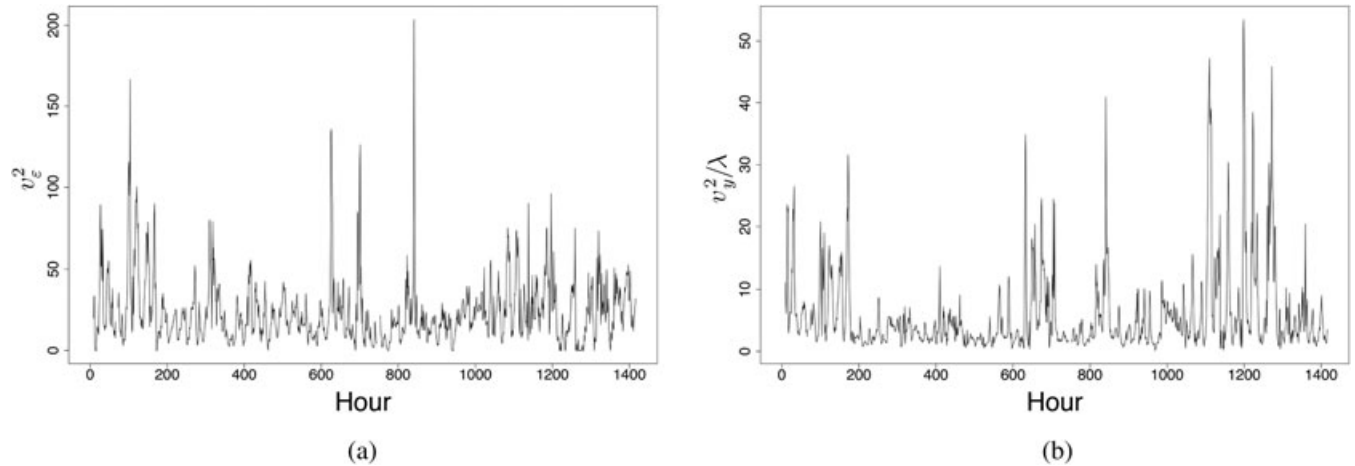


FIGURE 4 Time series plots of (a) $\hat{v}_\varepsilon^2$ and (b) $\hat{v}_\varepsilon^2 / \hat{\lambda}$

due to poor dispersion caused by meteorological and topographical effects in Southwestern Taiwan. The corresponding standard error map based on (4) is shown in Figure 2c. Not surprisingly, locations with fewer AirBoxes nearby tend to have higher standard error values.

Some interesting patterns can be tracked from the kriging maps. Three examples are given below. The first one is an important Taiwanese culture to worship the Jade Emperor during Chinese New Year's Eve (January 27, 2017) or Chinese New Year's Day (January 28, 2017). In Taiwan, people believe that Jade Emperor is the god who rules the universe and has

the capacity to bring good fortune. People worship the Jade Emperor by burning incense sticks and setting off firecrackers. The worship typically starts an hour before entering either day at 23:00. Because burning incense sticks and setting off firecrackers produce $\text{PM}_{2.5}$ particles, the $\text{PM}_{2.5}$ levels start to increase at 23:00 in the midwest and southwest of Taiwan before entering the Chinese New Year's Eve and reach a peak at 0:00 while entering Chinese New Year's Eve (see $\text{PM}_{2.5}$ maps in the second row of Figure 5). Interestingly, no such pattern can be seen in North Taiwan, where Taipei city is

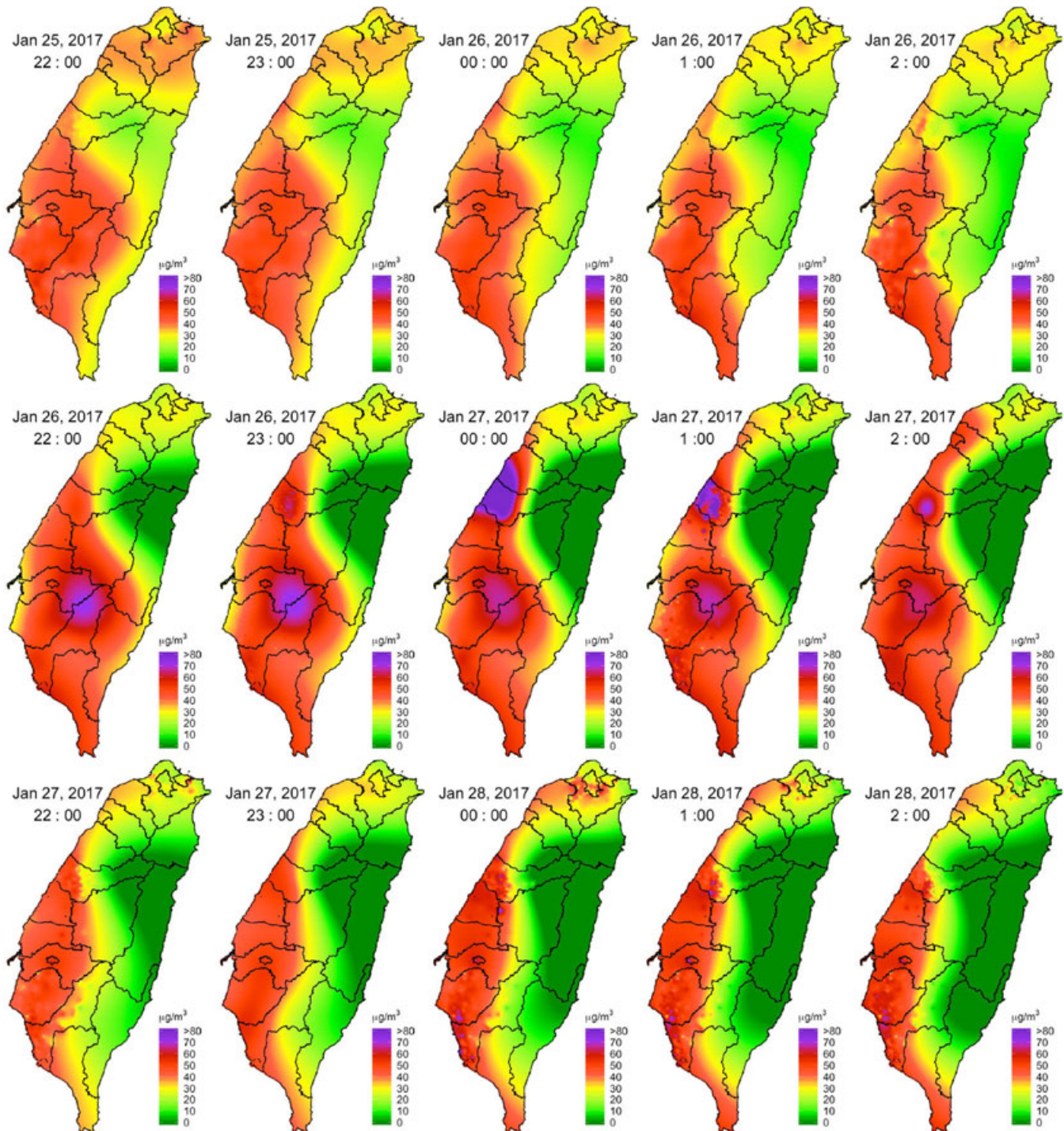


FIGURE 5 The event of Chinese New Year's Eve. $\text{PM}_{2.5}$ maps around Chinese New Year's Eve (January 27, 2017) and Chinese New Year's Day (January 28, 2018), where the first row shows some maps around the midnight of January 25 (a normal night), the second row shows some maps around the midnight of January 26 (with a strong increasing pattern in $\text{PM}_{2.5}$ levels), and the third row shows some maps around midnight of January 27 (with a weak increasing pattern in $\text{PM}_{2.5}$ levels). $\text{PM}_{2.5}$ = fine particulate matter

located, because firecrackers were banned in Taipei. Similar but less obvious results can be seen on the $PM_{2.5}$ maps in the third row of Figure 5 before and after entering the Chinese New Year's Day.

The second example happened at the Jade Emperor's birthday (the ninth day of the first month in the Chinese lunar calendar) on February 5, 2017. This is the most important day of the year to worship Jade Emperor. During the event, people burn incense, make food offerings, and light firecrackers. The activities typically start an hour before the day and end at about 9:00 on that day. These result in an increase of $PM_{2.5}$ levels throughout the period, which can be seen in the $PM_{2.5}$ maps on Figure 6. Midwest Taiwan can be seen to have more activities than other regions.

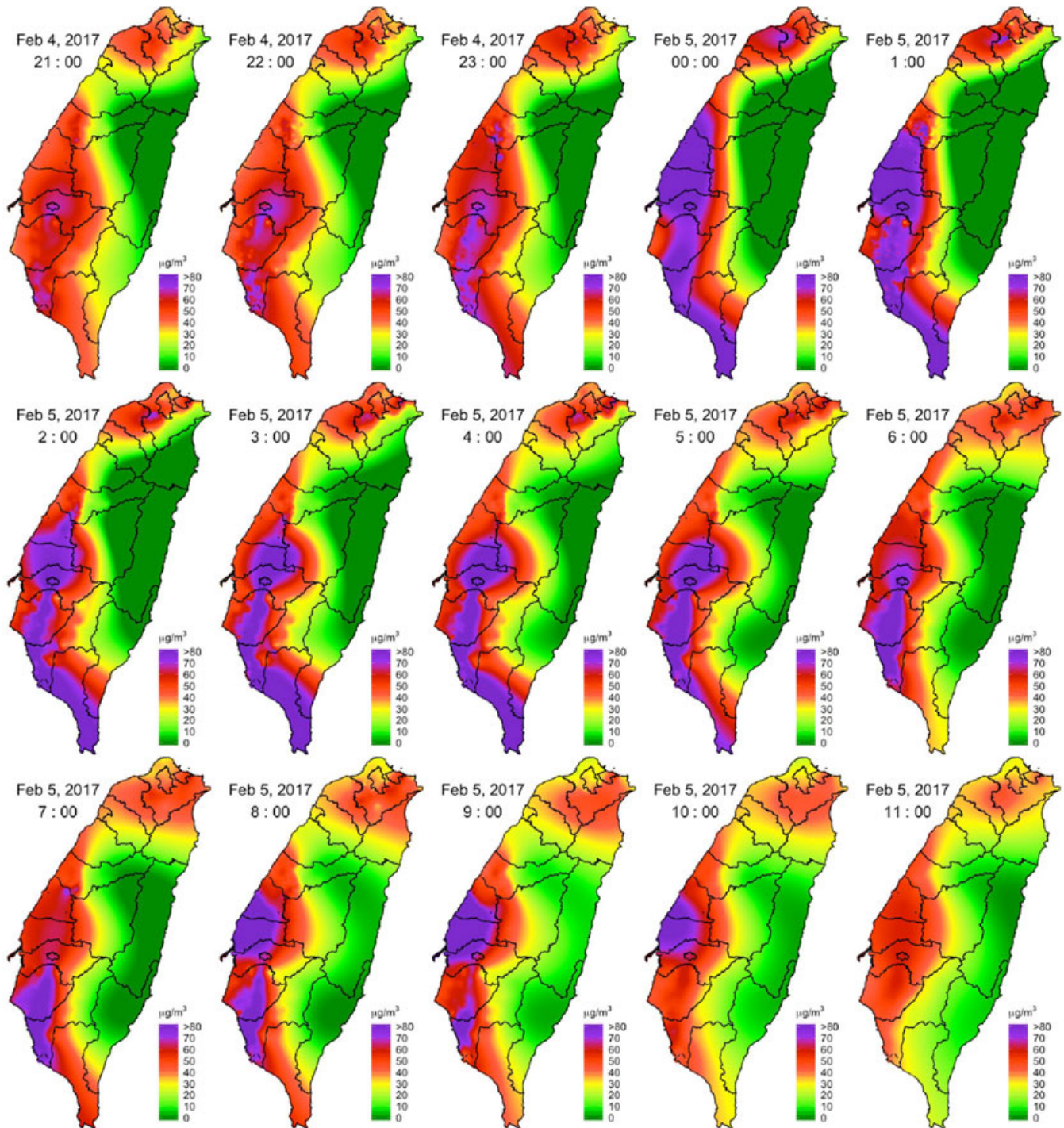


FIGURE 6 The event of Jade Emperor's birthday. $PM_{2.5}$ maps around Jade Emperor's birthday (February 5) in 2017, where the celebration starts around 23:00 on February 4 and ends around 9:00 on February 5. Midwest Taiwan can be seen to have more activities than other regions. $PM_{2.5}$ = fine particulate matter

The third example is the haze effect from Mainland China. Air pollution has become a major issue in Mainland China due to industrialization. Most of the big cities suffer serious haze of air pollution, including Beijing, Tianjin, and Shanghai. Because Taiwan lies at the southeast of Mainland China, haze can be transported from Mainland China to Taiwan during fall and winter when the Northeastern monsoon winds are strengthened by a high pressure system from China. For example, on February 10, 2017, PM_{2.5} levels (as shown on PM_{2.5} maps in the middle row of Figure 7) were high over the entire Taiwan due to haze and dust storms from Mainland China. These are in contrast to the corresponding maps on normal days, such as February 9 and 12 shown in the top and bottom rows of Figure 7.

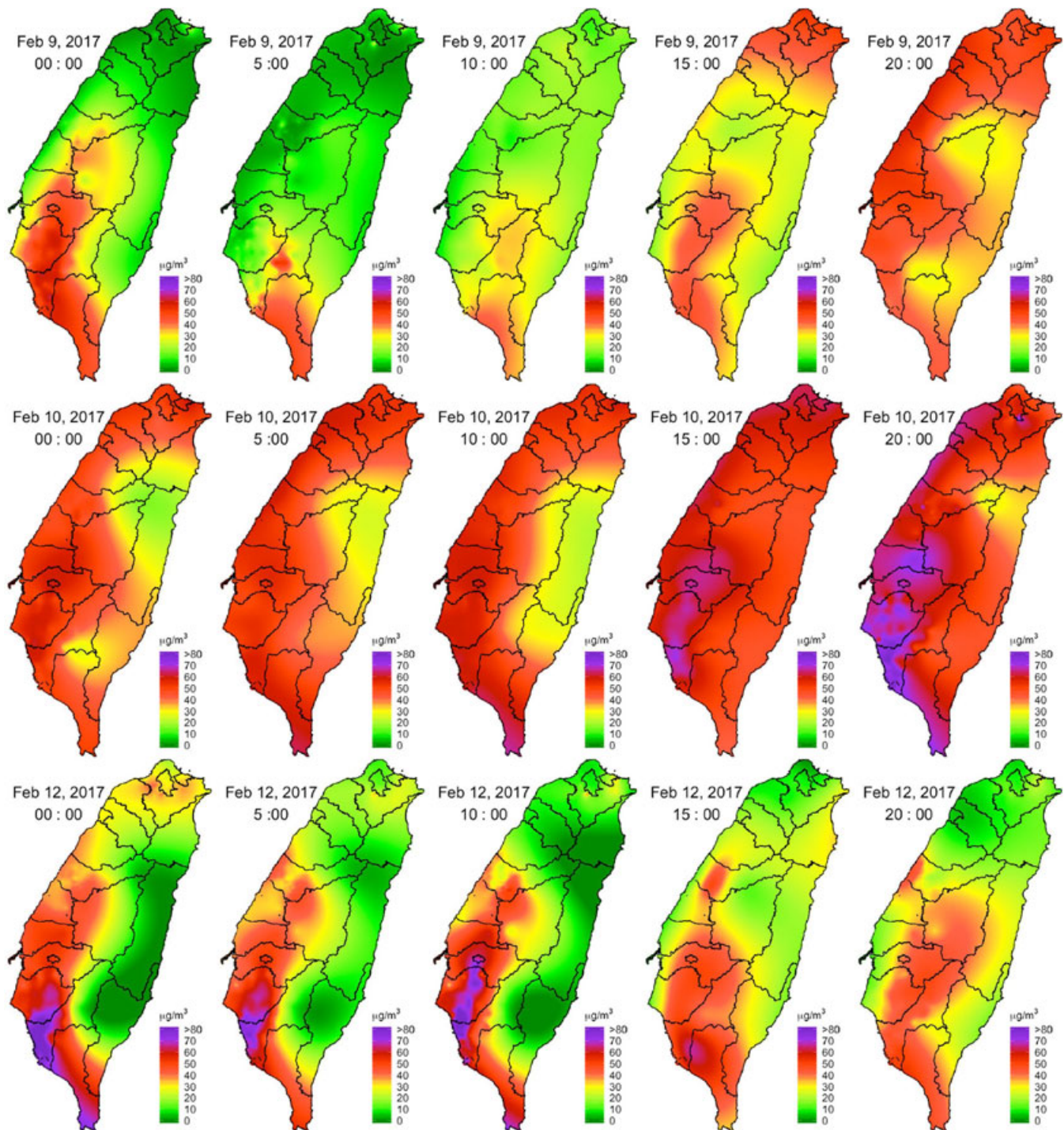


FIGURE 7 The event of a haze and dust storm from Mainland China. PM_{2.5} maps (top) before, (middle) during, and (bottom) after the storm, where the haze and dust storm starts to impact Taiwan at (top right) 20:00 on February 9. PM_{2.5} = fine particulate matter

4.2 | Anomalous measurements

Figure 2d displays the proposed standardized residuals of (5) at 8:00 on January 26, which show some unusual large or small $PM_{2.5}$ measurements. To identify unusual AirBoxes that tend to produce anomalous measurements, we rank the AirBoxes using the RMSE values of (6). The results are shown in Figure 8a. The corresponding biases of (7) are shown in Figure 8b. There are 85 AirBoxes with RMSE values larger than 3. We classify them into three groups according to $b_i \geq 3$, $b_i \in (-3, 3)$, and $b_i \leq -3$ (see Figure 8c), as discussed in Section 3.2.

Three control charts corresponding to AirBoxes randomly selected from each of the three groups with RMSE values greater than 3 are shown in Figure 9a–c. A randomly selected control chart with $RMSE_i \leq 3$ is shown in Figure 9d. As expected, the points in Figure 9a (with high RMSEs and high biases) tend to lie above the upper control line, which implies that the AirBox may be near a permanent emission source. The points in Figure 9b (with a high RMSE and a high negative bias) tend to lie below the lower control line, which implies that the AirBox may be indoors. The points in Figure 9c (with high a RMSE and a low bias) tend to have high variation, which implies that the AirBox may be abnormal. Finally, the points in Figure 9d (with a low RMSE) tend to lie between the two control limits, which implies that the AirBox is normal.

We also find some special features in the control charts. Figure 10 shows four examples. The two control charts shown on the top of Figure 10 likely correspond to two AirBoxes that are moved from outdoors to indoors at the time around hours 400 and 1,250 respectively, because their standardized residuals change from mostly in control to mostly fall below the lower control line. The two control charts in the bottom of Figure 10 show the opposite, which are likely to correspond to AirBoxes being moved from indoors to outdoors at the time around hours 1,200 and 625, respectively. The control chart in the lower-right panel has a change point at the time around hour 1,200, suggesting a possible new emission source added at the change point.

4.3 | A comparison with other methods

We compare the proposed kriging method described in Section 3.1 with two alternatives commonly used in producing maps for AirBoxes, namely, nearest neighbor regression (Altman, 1992) and the inverse distance weighting (IDW) method. For nearest neighbor regression, the $PM_{2.5}$ concentration at a location is estimated by the measurement from the AirBox

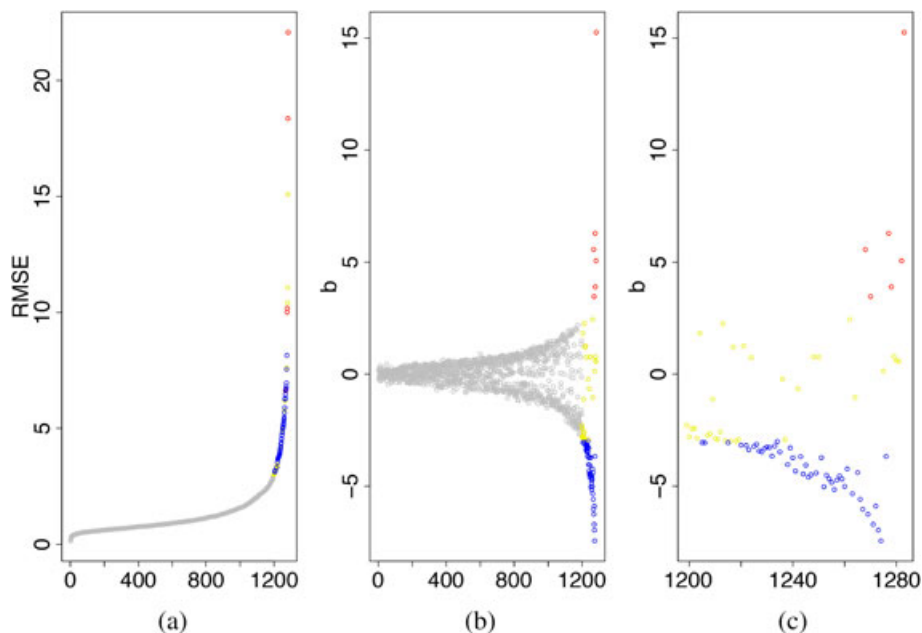


FIGURE 8 Identify the potential unusual AirBoxes according to the RMSE ($\mu g/m^3$): (a) ordered RMSE values for the AirBoxes; (b) the biases corresponding to RMSE values of (a); and (c) the biases corresponding to the 85 largest RMSE values with $RMSE_i > 3$, where six points with $b_i > 3$ are shown in red, 31 points with $b_i \in (-3, 3)$ are shown in yellow, and 48 points with $b_i < -3$ are shown in blue. RMSE = root mean squared error

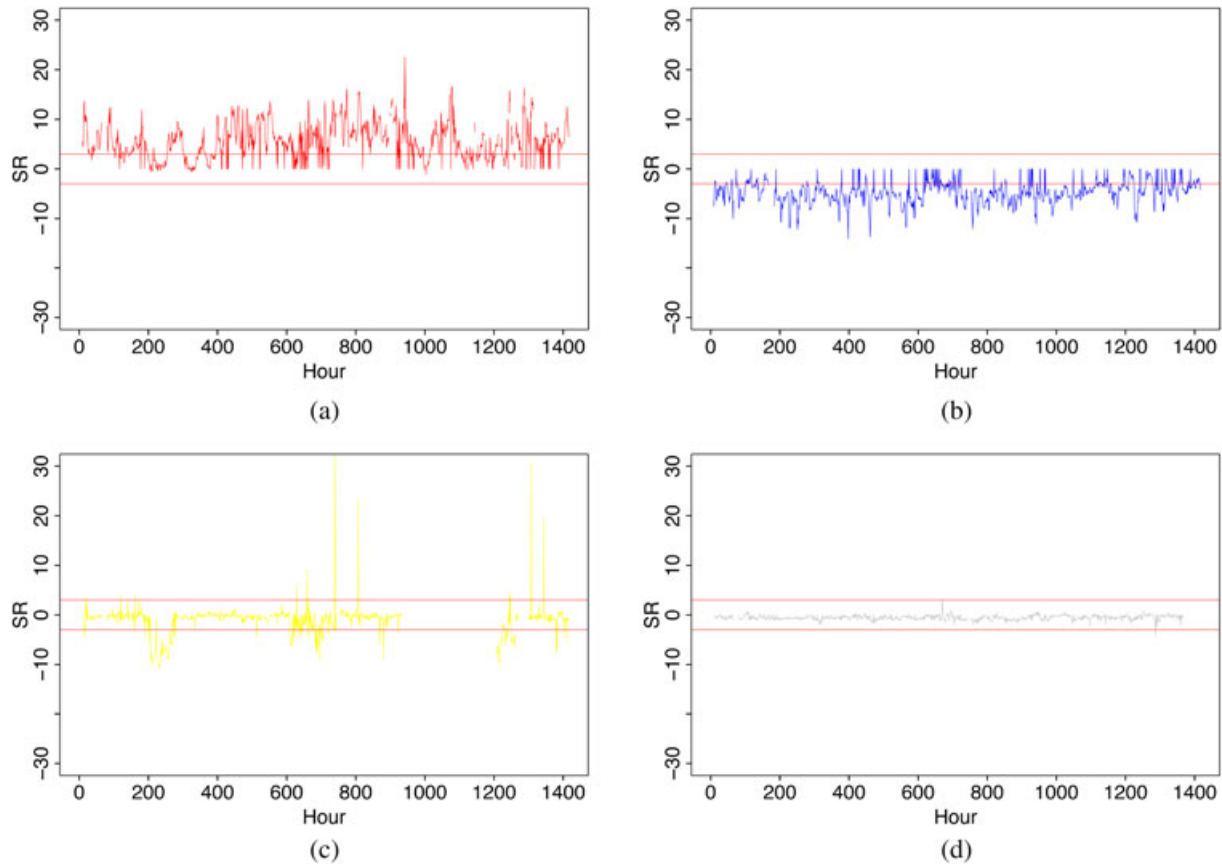


FIGURE 9 Four control charts corresponding to four randomly selected AirBoxes from the following four different classes: (a) $RMSE_i > 3$ and $b_i > 3$ (nearby a permanent emission source); (b) $RMSE_i > 3$ and $b_i < -3$ (indoor); (c) $RMSE_i > 3$ and $b_i \in (-3, 3)$, nearby a dining restaurant; and (d) $RMSE_i \leq 3$ (normal). RMSE = root mean squared error

that has the smallest distance to it. This produces a piecewise constant map partitioned by AirBox locations with their values given by the corresponding $PM_{2.5}$ concentrations. For IDW, the $PM_{2.5}$ concentration at location $\mathbf{s} \in D$ is estimated by a weighted average of all $PM_{2.5}$ observations, where the weights are given by the inverse distances to $\mathbf{s}$.

The three methods (nearest neighbor, IDW, and the proposed kriging) are compared using a test data set. For each hour, we randomly select 100 AirBoxes that have data observed, denoted by $z(\mathbf{s}_1^*), \dots, z(\mathbf{s}_{100}^*)$. We then compare the three methods according to the RMSE and the median absolute error (MAE) criteria as follows:

$$RMSE = \left\{ \frac{1}{100} \sum_{i=1}^{100} (\tilde{y}(\mathbf{s}_i^*) - z(\mathbf{s}_i^*))^2 \right\}^{1/2},$$

$$MAE = \text{median} \left\{ \left| \tilde{y}(\mathbf{s}_1^*) - z(\mathbf{s}_1^*) \right|, \dots, \left| \tilde{y}(\mathbf{s}_{100}^*) - z(\mathbf{s}_{100}^*) \right| \right\},$$

where $\tilde{y}(\mathbf{s}_i^*)$ is a generic estimate of $y(\mathbf{s}_i^*)$. Note that the MAE criterion is less affected by outliers. The results by averaging 24 hourly RMSE and MAE values over each of the 59 days in January and February 2017 for the three methods are shown in Figure 11. The proposed kriging method can be seen to produce much lower average RMSEs and much lower MAEs for almost all 59 days, indicating its effectiveness over the two commonly used methods. The averaged RMSE and MAE over 59 days are displayed in Table 1. The proposed kriging method reduces the error rates by 4.0% and 8.2% in terms of RMSE and MAE, respectively, over the IDW method. Those numbers increase to 21.6% and 21.0% over the nearest neighbor method.

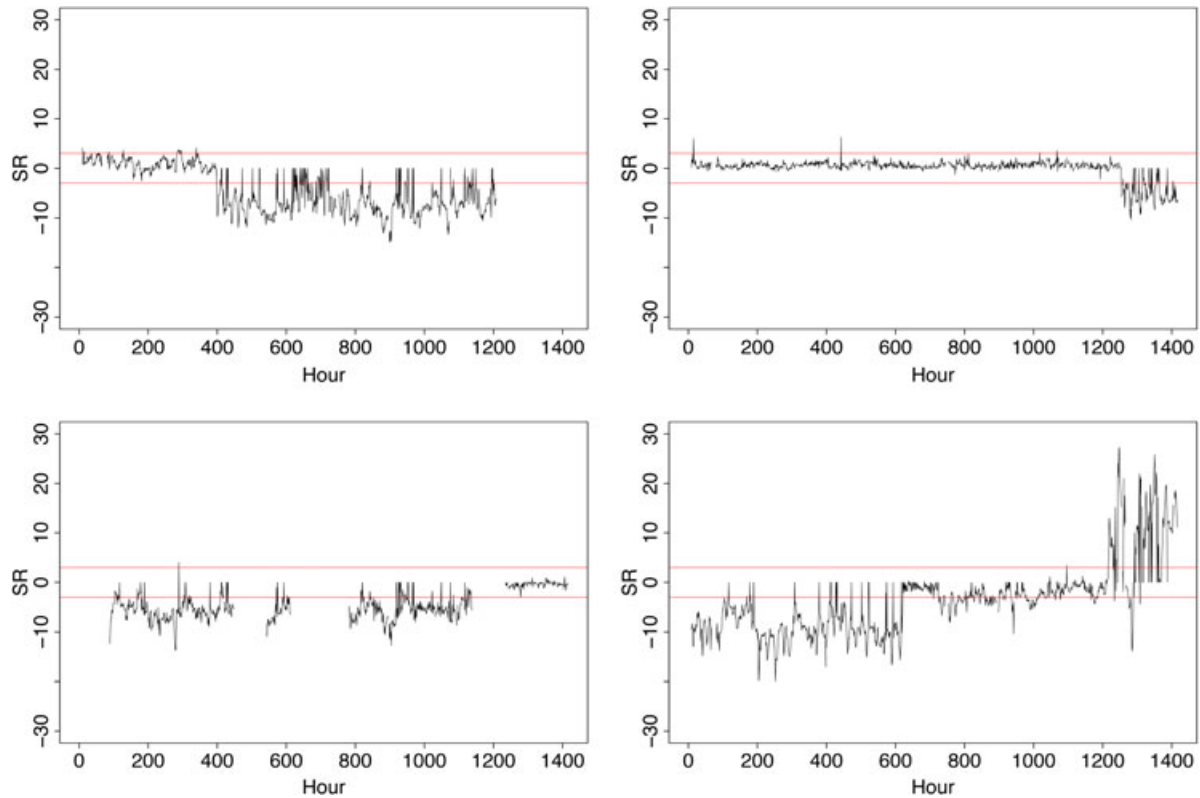


FIGURE 10 Control charts for some AirBoxes with special features

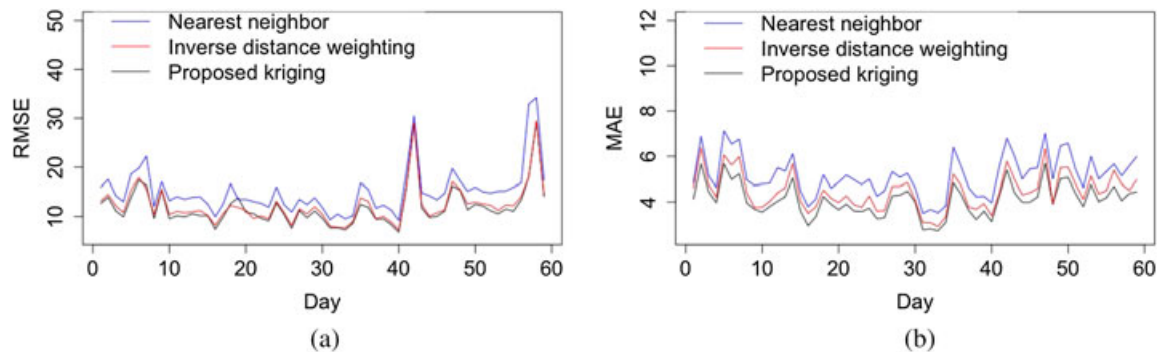


FIGURE 11 A comparison among nearest neighbor, inverse distance weighting, and the proposed kriging in terms of (a) RMSE ($\mu\text{g}/\text{m}^3$) and (b) MAE ($\mu\text{g}/\text{m}^3$). RMSE = root mean squared error; MAE = median absolute error

TABLE 1 The averaged RMSE and MAE for three methods ($\mu\text{g}/\text{m}^3$)

Method	Averaged RMSE	Averaged MAE
Nearest neighbor	15.16	5.18
Inverse distance weighting	12.37	4.46
The proposed kriging	11.88	4.09

Note. RMSE = root mean squared error; MAE = median absolute error.

5 | CONCLUSIONS AND DISCUSSION

The AirBox project launched in Taiwan provides a new and effective method for monitoring $\text{PM}_{2.5}$. With more and more AirBoxes being deployed, we have a better ability to obtain more precise $\text{PM}_{2.5}$ measurements at finer and finer spatial and temporal resolutions, which may eventually allow us to pinpoint the exact location of an emission source.

In this paper, we proposed to use kriging with multiresolution spline basis functions, which provides real-time PM_{2.5}-level estimation at any location in Taiwan with uncertainty quantification. We also proposed a new method to detect anomalies and developed a spatio-temporal control chart to monitor whether the PM_{2.5} data produced from AirBoxes are reliable. Our method is able to detect potential emission sources, malfunctioned AirBoxes, and AirBoxes that are put indoors.

Regarding the computation, the proposed kriging method is computationally fast when n is not very large even if the n -dimensional matrix inversion in (3) is taken directly. It takes about 30 s to obtain $\hat{\beta}$, $\hat{\theta}$, and high-resolution maps in Figures 2b and 2c on a personal computer (with Intel Core i7 3.4 GHz) using R. For a very large n , we can obtain a good approximation of (3) and (4) using the R package LatticeKrig (Nychka, Hammerling, Lindgren, Sain, & Lenssen, 2016).

Although AirBoxes provide high-resolution spatial and temporal coverage, the measurements from these small IoT devices have much higher measurement errors than those from large monitoring stations. Their accuracies may also be affected by temperature and humidity. Further research is needed to incorporate both types of data. In addition, it is of interest to forecast PM_{2.5}, which may require using a computer model, a statistical model, or a combination of the two. We consider this an important topic for our future research.

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